

湖南文理学院

HUAS-Beamer Template Demo

— A Gallery of Layout Elements for Academic Talks

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1 Text and Lists

2 Layout and Figures

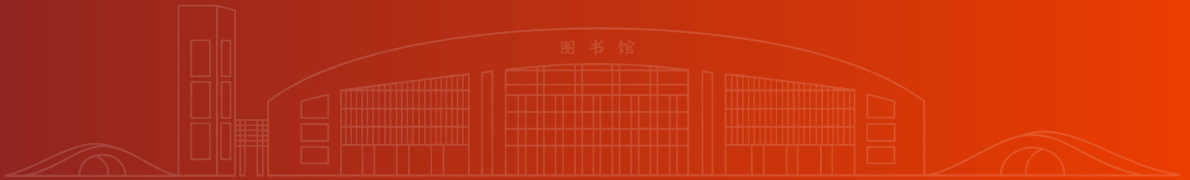
3 Blocks, Theorems and Formulas

4 Tables, Code and Summary

Guide: each slide demonstrates one category of layout elements, with neutral sample text from academic-report settings.

Section 1

Text and Lists



Contents

1 **Text and Lists**
- Multilevel Lists
- Description and Emphasis

2 Layout and Figures

3 Blocks, Theorems and Formulas

4 Tables, Code and Summary

Multilevel Lists: itemize and enumerate

- Main research contents
  - Data collection and preprocessing: build the experimental dataset;
  - Model design and training: compare three baseline methods
    - Feature extraction and hyper-parameter settings;
    - Training, validation and ablation studies.
  - Result analysis and technical report writing.
- Highlight: milestone deliverables and progress control.

1. Phase one: literature review and proposal design;
2. Phase two: system implementation and experiments
   1. Finish module development and unit testing;
   2. Organise experimental data and interim summary.

Description Lists and Emphasis

Survey Literature review and requirement analysis;

Model Model construction and parameter estimation;

Experiments Comparative studies and result analysis;

Evaluation Metric statistics and error analysis;

Summary Conclusion drawing and report writing.

Ways of emphasis:

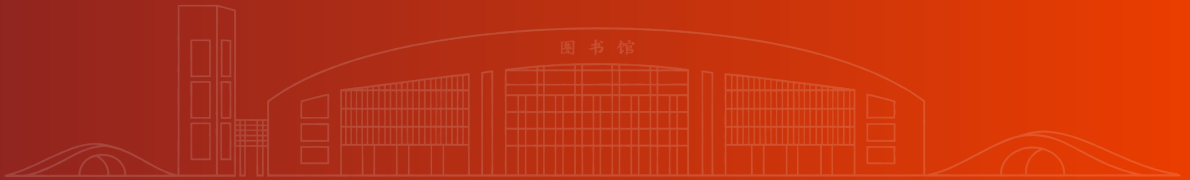
- **alert bold**: eye-catching notice;
- **bold**: highlight key terms;
- *italic*: mark technical notions;
- theme colour: match the palette.

Note: the sample text on this slide is for layout only;replace it page by page with your own[1].

[1] Footnote example: notes are placed at the bottom of the page in the theme colours.

Section 2

Layout and Figures



Contents

1 Text and Lists

2 Layout and Figures
- Column Layout
- Figures and Captions
- Side-by-side Figures

3 Blocks, Theorems and Formulas

4 Tables, Code and Summary

Column Layout: Two and Three Columns

Completed work

- Dataset construction and baseline reproduction;
- Comparative experiments and preliminary analysis.

Future work and schedule

- Additional ablation studies and error analysis;
- System prototype and paper writing.

Problem
Baselines underfit in small-sample settings.

Solution
Data augmentation with regularisation.

Evaluation
Cross-validation and significance tests.

Figures and Captions

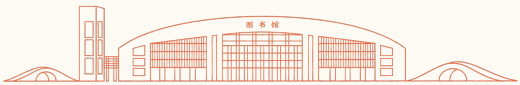


Figure 1: Line-art campus building

- Captions are numbered automatically in theme colours;
- Figures mix freely with text in columns;
- Assets are kept in the `figures/` folder;
- Cross-reference in text: see Fig. 1.

Side-by-side Figures and Cross-references



Figure 2: Side-by-side figure 1 (line art)

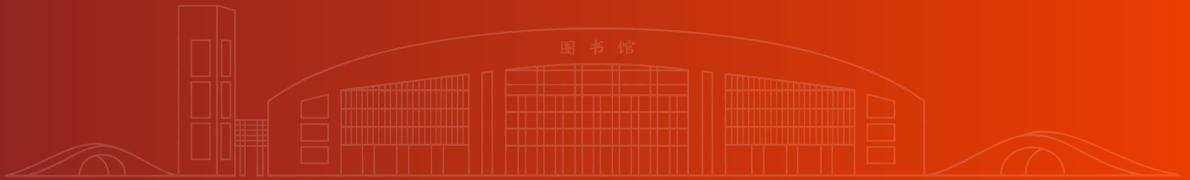


Figure 3: Side-by-side figure 2 (badge)

Fig. 2 and Fig. 3 sit side by side in one row; cross-references update automatically as floats are added or removed, **with no manual maintenance**.

Section 3

Blocks, Theorems and Formulas



Contents

1 Text and Lists

2 Layout and Figures

3 Blocks, Theorems and Formulas

- Block Environments
- Theorems and Proofs
- Formulas

4 Tables, Code and Summary

The Three Block Environments

Work summary (block)

Dataset construction, reproduction of three baselines and design of the comparative study are completed; an interim report is available.

Result example (exampleblock)

On the public dataset, the proposed method improves accuracy by 3.2 percentage points over the best baseline.

Risks and remedies (alertblock)

Tight schedule: keep a two-week buffer; if data are insufficient, switch to a public benchmark dataset.

Theorem, Proof and Example

Theorem (Pythagorean theorem)

In a right-angled triangle, the sum of the squares of the two legs equals the square of the hypotenuse.

Proof.

Let the legs be a, b and the hypotenuse c. Arrange four congruent copies of the triangle into a square of side $a + b$; comparing the two area decompositions yields $a^2 + b^2 = c^2$. □

Example

For $a = 3$ and $b = 4$ we get $c = \sqrt{a^2 + b^2} = 5$.

Inline and Display Equations

Inline math: the objective is $J(\theta)$, the learning rate is $\alpha = 0.05$, and the sample size is $n = 320$.

$$k : [-\pi, \pi] \to [0, 1]$$

Numbered equation: the network input is an image tensor

$$\mathrm{x} \in \mathbb{R}^{C_{\text{in}} \times H \times W} \tag{1}$$

where C_{in} is the number of channels, H the height and W the width; Eq. (1) determines the tensor shapes of each layer.

Multiline Equations and Cases

Aligned equations (align):

$$Q_{\text{target}} = r + \gamma\, Q^{\pi}(s', \pi_\theta(s')) + \epsilon, \tag{2}$$

$$\epsilon \sim \operatorname{clip}\big(\mathcal{N}(0, \sigma), -c, c\big). \tag{3}$$

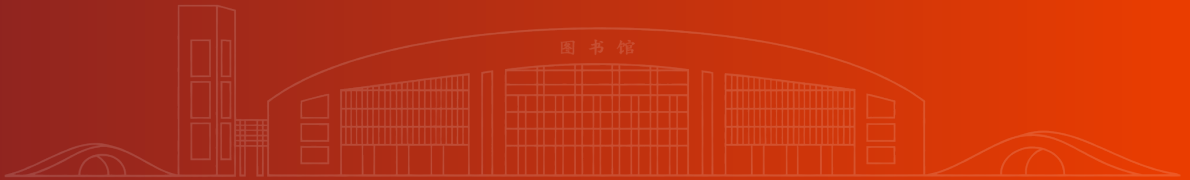
Piecewise function (cases):

$$L(e) = \begin{cases} e^2, & |e| \le 1; \\ 2|e| - 1, & |e| > 1. \end{cases}$$

Here r is the immediate reward, γ the discount factor and Q^{π} the action-value function; $L(e)$ is a robust loss that varies piecewise with the error e.

Section 4

Tables, Code and Summary



Contents

1 Text and Lists

2 Layout and Figures

3 Blocks, Theorems and Formulas

4 Tables, Code and Summary

- Tables and Code
- Citations and Navigation
- Summary and Outlook

Tables with Coloured Headers

Table 1: Unique-value counts per field (sample)

Field	# unique values
posting_id	34250
image	32412
image_phash	28735
title	33117
label_group	11014

Tab. 1 uses `booktabs` rules together with the `\huastableheader` macro; row spacing is set centrally by the template, and `multirow`/`multicolumn` merge cells.

Code Listings

```latex
\begin{itemize}
  \item Data collection and preprocessing
  \item Model design and training
  \begin{itemize}
    \item Feature extraction and tuning
  \end{itemize}
\end{itemize}
```

Rendered result:

- Data collection and preprocessing
- Model design and training
  - Feature extraction and tuning

Keywords, comments and strings are coloured separately; line numbers appear on the left, and long lines wrap automatically (`breaklines`).

Citations and Footnotes

Citation examples

- Single citation: a survey of data-driven methods[4];
- Multiple citations: training methods and typesetting tools[1, 3];
- Citations often go with footnotes to trace data sources and remarks[5, 2].

Quotation and footnote

> *"Good tools are a prerequisite to the successful execution of a job."*[2]

References are managed with `bibLaTeX` and numbered automatically; full entries appear on the References pages (`allowframebreaks` splits them if needed).

[2] Quotation example: footnotes are placed at the page bottom.

Hyperlinks and Navigation

`url` `https://www.huas.cn/`

`href` Hunan University of Arts and Science

`jump` Jump to the element gallery

- Cross-references: Fig. 1, Tab. 1, Eq. (1);
- Bottom-right: navigation buttons and page numbers;
- TOC entries and bibliography items are clickable.

Element Gallery

Table 2: Layout elements covered by this demo

Category	Environments/commands	Example pages
Lists	itemize / enumerate / description	Multilevel and description lists
Layout	columns / figure / caption	Columns, text and figures
Blocks	block / exampleblock / alertblock	The three block environments
Theorems	theorem / proof / example	Theorem and proof
Math	equation / align / cases	Inline, display and multiline
Tables	tabular + huastableheader	Coloured header rules
Code	lstlisting	Code listings
Refs	cite / footnote / hyperlink	Citations and navigation

Closing Remarks

What the template gives you

- Consistent colours, fonts, headers, footers and decorations;
- All common layout elements are pre-styled and ready to use.

Suggestions

- Write your content first, then apply the layout; any slide here can be copied and rewritten;
- For fine tuning, search the starred comments in `huasbeamer.sty`;
- The cool blue-white scheme only needs the `blue` theme option.

References I

[1] Hao Chen and Fang Wang.
Robust model training with limited samples.
In *Proceedings of the International Conference on Artificial Intelligence and Applications*, pages 123–130, 2022.

[2] Lei Gao.
A benchmark suite for academic presentation analysis.
Technical Report TR-2024-01, Data Systems Laboratory, 2024.

[3] Qiang Li.
Academic Typesetting with LaTeX.
Modern Education Press, Beijing, 2020.

References II

[4] Wei Zhang and Ming Li.
A comparative study of deep learning methods for tabular data.
Journal of Data Science and Applications, 12(2):88–101, 2023.

[5] 王静.
演示文稿自动评估方法研究.
Master's thesis, 示例大学, 2021.

Thank You!

Q & A

Hunan University of Arts and Science

